# Vendor Change Ticket Bundle: Rollout 2025-11-20 to 2025-11-21

Category: Reference | Model: VFIP-CT-ROLL-1120

Description: Compiled change ticket summaries, vendor release notes, O&M incident report, dispatch runbooks, and settlement guidance related to the XG-440 firmware incident and associated firmware rollout (v3.14.2 → v3.15.0). Includes acceptance checklists, verification procedures, rollback triggers, and escalation notes for RAG use.

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## 1. Overview & Purpose

This document consolidates vendor-supplied change tickets and related artifacts for the firmware rollout executed across our portfolio on 2025-11-20 and 2025-11-21 (model XG-440). It is intended to serve as a canonical reference for operations, engineering, and knowledge-assistant systems performing retrieval-augmented generation (RAG).

Included materials:

- Vendor release notes for firmware v3.14.2 through v3.15.0
- Change ticket summaries, acceptance criteria, rollbacks, and staged rollout phases (A–D)
- O&M incident report and postmortem timeline tied to the incident
- Dispatch runbooks mapping SCADA alarm codes to CMMS corrective work orders
- ERCOT settlement and imbalance charge reconciliation guidance
- Contact and escalation points, and time-stamped ticket comments correlated to site observations

Scope: This package is synthetic and designed for internal operational reference and knowledge-base ingestion. All codes, serial ranges, and examples are presented for operational fidelity and RAG downstream use.

## 2. Vendor Firmware Release Notes (v3.14.2 → v3.15.0)

Summary of changes across releases included in the rollout. This section highlights parameter changes, protective trip updates, and known behavior changes that informed rollback decisions.

### 2.1 Parameter Changes & Default Values

Table: Key parameter changes introduced between versions v3.14.2 and v3.15.0.

| Parameter | Old Default (v3.14.2 | New Default (v3.15.0) | Impact / Notes |
|---|---|---|---|
| protection.trip_threshold.trip31_pct | 120 | 110 | Lowered threshold to reduce false-negative detection of overcurrent events. May increase trips under transient grid conditions. |

| | | | |
|---|---|---|---|
| protection.retry.count | 2 | 3 | Increases retry attempts before permanent lockout for comms failures. |
| firmware_update.concurrent_sessions | 10 | 6 | Reduces simultaneous upgrade load to lower device memory pressure. |
| scada_alarm_code.baseline_window_minutes | 60 | 120 | Longer baseline window for alarm trending to reduce transient alerts during rollout. |
| logging.level | INFO | DEBUG (selective modules) | Temporary increase for diagnostics; may increase storage usage. |

**Parameter Example — JSON snippet**

```
{
  "protection": {
    "trip_threshold": { "trip31_pct": 110 },
    "retry": { "count": 3 }
  },
  "firmware_update": { "concurrent_sessions": 6 },
  "scada_alarm_code": { "baseline_window_minutes": 120 },
  "logging": { "level": "DEBUG", "modules": ["comm", "protection"]
}
}
```

Operational note: Any site-level overrides previously applied must be audited and re-applied after upgrade if necessary. Use "device_config.override" logs to verify persistence.

### 2.2 Protective Trip Codes & Semantics

Primary trip codes referenced in vendor documentation and tickets:

| Trip Code | Description | Symptoms | Recommended Response |
|---|---|---|---|
| TRIP-31 | Overcurrent instantaneous protective trip | Immediate inverter shutdown; scada_event: "trip:31"; field: protection.last_trip_code = 31 | Investigate grid fault; check RMS current logs; if repeated post-upgrade (>3 occurrences/site in 24h), trigger rollback per vendor guidelines. |
| TRIP-44 | DC string mismatch / arc detection | Inverter FAULT state; scada_alarm_code: 44; string-level error logs present | Isolate affected string, validate insulation resistance, escalate to vendor for hardware verification. |
| TRIP-02 | Grid frequency excursion lockout | site unavailable; scada_event includes freq_hz outside [58.5, 60.5] | Coordinate with ISO/TO; confirm grid conditions before concluding firmware responsibility. |

Important: The vendor updated TRIP-31 behavior in v3.15.0 to be more aggressive at lower trip31_pct thresholds (see parameter changes). This directly influenced rollback criteria used during the rollout.

### 2.3 Recommended Rollback Conditions

Vendor-defined rollback triggers included in tickets:

- Portfolio availability decline > 2% within 24 hours of rollout start — immediate rollback recommended.
- Repeated TRIP-31 occurrences: > 3 unique TRIP-31 events per site within 24 hours, or > 5 occurrences across the fleet correlated to rollout wave — rollback.
- Critical SCADA alarm increase: scada_alarm_code baseline deviation > +200% compared to pre-rollout baseline for 2 consecutive 1-hour windows.
- Unresolvable communication lockups: > 2% devices in "firmware_update.failed" state after retry.count attempts, or devices stuck in "reboot_loop".

Rollback procedure summary (high-level):

1. Stop further staged waves; place remaining batches in hold.
2. Trigger vendor rollback toolset to re-image affected devices to v3.14.2 (or previously validated image).

3. Execute post-rollback verification (See Section 4).
4. Open incident ticket and notify ERCOT dispatch if availability impacts cross contingency limits.

## 3. Change Ticket Summaries — Rollout & Hotfix

Each ticket below is a vendor portal summary extracted and normalized for operational use. Tickets were executed in staged batches A–D.

### 3.1 Ticket Template & Fields

Standard ticket fields used across the vendor portal export:

| Field | Description / Format |
|---|---|
| Ticket ID | Unique vendor ID (e.g., VEND-2025-11120-001) |
| Change Window | UTC start/end (e.g., 2025-11-20T00:00Z to 2025-11-20T06:00Z) |
| Targeted Plant List | Plant names and IDs; often grouped by geographic region |
| Inverter Serial Ranges | Range notation (e.g., XG440-1000001 → XG440-1001250) |
| Firmware Mapping | From vX.Y.Z to vA.B.C; may include hotfix tag |
| Rollout Phase | A, B, C, or D (staged batches) |
| Acceptance Checklist | Predefined steps (ping, firmware_version, scada_alarm_code baseline, etc.) |
| Rollback Triggers | Vendor-defined triggers and thresholds |
| Contacts | Vendor L1/L2, internal ops, escalation phone/email |
| Comments / Time-stamped Logs | Portal comments with timestamps correlated to site events |

### 3.2 Tickets: 2025-11-20 to 2025-11-21

Representative set of change tickets executed during the main rollout. Each entry is an extracted summary suitable for RAG indexing.

**Ticket: VEND-2025-11120-001 — Batch A (2025-11-20 00:00Z → 2025-11-20 06:00Z)**

| | |
|---|---|
| **Target Plants** | North Ridge (PLANT-NR-001), LakeView (PLANT-LV-004) |
| **Inverter Serial Ranges** | XG440-1000001 → XG440-1000500 |
| **From → To** | v3.14.2 → v3.15.0 |
| **Phase** | A |
| **Acceptance Checklist (summary)** | 1. Pre-check: ping device_ip (3x) — expected RTT < 200ms<br>2. Verify current firmware_version == "3.14.2"<br>3. Baseline scada_alarm_code counts over previous 120 minutes<br>4. Confirm site availability > 98% for preceding 24h<br>5. Ensure firmware_update.concurrent_sessions limit set to 6 |
| **Rollback Triggers** | >2% portfolio availability decline; >3 TRIP-31 events/site in 24h; scada_alarm_code baseline >+200% |
| **Contacts** | Vendor L1: vendor-l1@vendor.com, L2: vendor-l2@vendor.com, Ops: ops-oncall@company.com, Escalation: vendor-escalate@vendor.com |
| **Ticket Comments (selected)** | 2025-11-20T00:12Z — vendor: "Batch A started, 120 devices currently under update."<br>2025-11-20T01:05Z — ops: "2 devices reported firmware_update.failed; vendor retry invoked (retry.count=3)"<br>2025-11-20T03:10Z — ops: "scada_alarm_code increase: +15% vs baseline; monitoring."<br>2025-11-20T05:00Z — vendor: "Batch A completed; no TRIP-31 spike observed; moving to Batch B." |

**Ticket: VEND-2025-11120-002 — Batch B (2025-11-20 08:00Z → 2025-11-20 14:00Z)**

| | |
|---|---|
| **Target Plants** | Prairie East (PLANT-PE-007), Riverbend (PLANT-RB-002) |
| **Inverter Serial Ranges** | XG440-1000501 → XG440-1001000 |
| **From → To** | v3.14.2 → v3.15.0 |
| **Phase** | B |

| | |
|---|---|
| **Acceptance Checklist (summary)** | 1. Pre-upgrade ping & access checks<br>2. Verify config overrides saved (device_config.override)<br>3. Check inverters in last 24h have no TRIP-31 occurrences |
| **Rollback Triggers** | Same as Ticket 001; additional trigger: >2% devices in "reboot_loop" |
| **Comments (selected)** | 2025-11-20T08:05Z — vendor: "Batch B started."<br>2025-11-20T10:25Z — ops: "Riverbend reported two TRIP-31 events at 10:12Z and 10:18Z on serial XG440-1000723."<br>2025-11-20T10:40Z — vendor-l2: "Investigating logs; requesting power curve data for 1000723."<br>2025-11-20T11:30Z — ops: "TRIP-31 events resolved after device reboot; monitoring for recurrence."<br>2025-11-20T13:50Z — vendor: "Batch B completed; no rollback required." |

**Ticket: VEND-2025-11121-003 — Batch C → D (2025-11-21 00:00Z → 2025-11-21 23:50Z)**

| | |
|---|---|
| **Target Plants** | Suncrest (PLANT-SC-010), Canyon (PLANT-CN-011), Summit (PLANT-SU-012) |
| **Inverter Serial Ranges** | XG440-1001001 → XG440-1002000 |
| **From → To** | v3.14.2 → v3.15.0 |
| **Phase** | C & D (C early morning; D late evening) |
| **Acceptance Checklist** | 1. Confirm SCADA baseline for 120 minutes.<br>2. Ensure remote reboot capability verified for each in-field controller.<br>3. Confirm spare comms Gateway onsite at Suncrest. |
| **Rollback Triggers** | Escalation to immediate rollback if portfolio availability decline >1.5% within first 6 hours for these regions (higher sensitivity due to constrained grid conditions) |

| | |
|---|---|
| **Comments (selected)** | 2025-11-21T02:10Z — ops: "Suncrest reported multiple TRIP-31 events across three inverters; vendor notified."<br>2025-11-21T04:20Z — vendor-l2: "Analysis shows overcurrent spikes coincident with utility-side capacitor switching event at 04:18Z; recommend hold."<br>2025-11-21T06:00Z — ops: "Decision: HOLD batch D pending further analysis."<br>2025-11-21T18:45Z — vendor: "Batch D resumed with reduced concurrency; no further TRIP-31 clusters." |

### 3.3 Hotfix: 2025-12-08 (Post-rollout remediation)

After initial rollout, vendor issued a hotfix and targeted change tickets to address edge cases.

| | |
|---|---|
| **Ticket** | VEND-2025-12008-HOTFIX-01 |
| **Change Window** | 2025-12-08T02:00Z → 2025-12-08T08:00Z |
| **Scope** | Devices that reported TRIP-31 events during rollout (approx. 325 devices). Serial set exported via vendor query and validated. |
| **Action** | Apply patch v3.15.0-hf1 (addressing nominal overcurrent sensing calibration) and adjust protection.trip_threshold.trip31_pct to 112 for affected devices. |
| **Rollback** | Re-image to v3.14.2 if TRIP-31 repeats more than 2 times per device within 48 hours of hotfix. |
| **Selected Comments** | 2025-12-08T02:05Z — vendor: "Hotfix rollout started; targeted set=325 devices."<br>2025-12-08T04:30Z — ops: "50 devices reported 'firmware_update.pending_reboot'; vendor initiating automated reboot." |

## 4. Acceptance, Pre- and Post-Upgrade Verification

Detailed procedural steps to verify device state before and after upgrades. Use these as canonical checks for automation and runbooks.

### 4.1 Prerequisites

- Network level: Ensure device management VLAN reachable; test with 3x ping <200ms.
- Data retention: Export device_config and power_curves for the last 24 hours to object store.
- Grid conditions: Confirm site frequency > 59.5 Hz and < 60.5 Hz for previous 30 minutes.
- Staffing: Ops on-call + vendor L2 present during waves B and D.

### 4.2 Pre-upgrade Steps (detailed)

1. Run device discovery query:

```
# Example query (pseudo-CLI)
GET /devices?model=XG-440&fw_version=3.14.2&status=online
```

2. For each device:
    1. Ping device_ip 3 times: expected RTT < 200ms.
    2. Verify firmware_version field:

    ```
    device.firmware_version == "3.14.2"
    ```

    3. Capture scada_alarm_code baseline for previous 120 minutes and store as baseline_.json.
3. Confirm no active TRIP-31 events in last 24 hours:

```
count(events where event.code == 31 and event.ts > now() -
24h) == 0
```

4. Set maintenance window and set device to "maintenance" mode in CMMS.

### 4.3 Upgrade Execution Steps (detailed)

1. Start vendor upgrade job with concurrency parameter set to 6:

```
vendor.upgrade.start({
  "devices": [...],
  "target_firmware": "v3.15.0",
  "concurrent_sessions": 6
});
```

2. Monitor per-device status: expected progress transitions: queued → downloading → installing → rebooting → online/healthy
3. If any device reports "firmware_update.failed", trigger automated retry (up to retry.count=3). If still failing, record device and continue monitoring.

### 4.4 Post-upgrade Verification

Post-upgrade checks must be performed at t+15m, t+60m, and t+24h.

1. Verify firmware_version = "3.15.0" for all updated devices:

```
SELECT device_id FROM devices WHERE firmware_version !=
"3.15.0";
```

2. Ping health check:

```
for device in devices: ping(device.ip) → expect RTT < 400ms
and no packet loss
```

3. Query protection.last_trip_code and protection.last_trip_ts to surface any TRIP-31 events.

```
SELECT device_id, protection.last_trip_code,
protection.last_trip_ts FROM devices WHERE
protection.last_trip_ts > now()-24h;
```

4. Compare scada_alarm_code counts vs pre-upgrade baseline:

```
alarm_delta_pct = (post_count - baseline_count) /
baseline_count * 100
```

   Trigger alerts if alarm_delta_pct > +200% for two consecutive 1-hour windows.

5. Availability check:

```
fleet_availability = sum(device.available_minutes_last_24h)
/ (number_of_devices * 1440) * 100
```

   Rollback if fleet_availability drops >2% vs pre-rollout baseline within first 24 hours.

6. Log and archive all diagnostic bundles for affected devices (include power curves from t-2h to t+2h).

### 4.5 Troubleshooting Flow (summary)

1. If single-device failure: collect diagnostics → attempt local reboot → vendor remote debug.
2. If cluster event at a plant (multiple TRIP-31): investigate grid events (TO/ISO), confirm correlation, sample power curve to identify overcurrent timing.
3. If cross-fleet impact observed (availability > thresholds): halt rollout, activate rollback per Section 2.3.

## 5. O&M Incident Report & Postmortem (XG-440)

Summary of incident tied to the rollout and the ensuing postmortem. This section is arranged as a timeline, root-cause hypothesis, corrective actions, and action items.

### 5.1 Executive Summary

Between 2025-11-20 and 2025-11-21, a firmware rollout to v3.15.0 impacted several sites with increased TRIP-31 protective trips and a modest availability decline. Vendor and internal teams performed targeted hotfixes and rollbacks; a hotfix was applied on 2025-12-08. No safety incidents occurred; financial impacts included curtailment windows and potential imbalance charges (see Section 7).

### 5.2 Timeline (Day-by-day)

| Timestamp (UTC) | Event |
|---|---|
| 2025-11-20T00:00Z | Batch A rollout begins (VEND-2025-11120-001) |
| 2025-11-20T01:05Z | Two devices report firmware_update.failed; vendor retries |
| 2025-11-20T08:00Z | Batch B begins; Riverbend reports initial TRIP-31 events |
| 2025-11-20T11:30Z | Batch B completes; no rollback required |
| 2025-11-21T02:10Z | Suncrest reports multiple TRIP-31 events across several inverters |
| 2025-11-21T04:20Z | Vendor correlates TRIP-31 to utility-side capacitor switching; Batch D placed on HOLD |
| 2025-11-21T18:45Z | Batch D resumed with reduced concurrency |
| 2025-12-08T02:00Z | Vendor hotfix rollout (v3.15.0-hf1) to 325 affected devices |
| 2025-12-09T10:00Z | Hotfix audit completed; TRIP-31 recurrence reduced by 92% in affected set |

### 5.3 Root Cause Analysis (RCA)

Findings:

1. v3.15.0 changed trip31_pct from 120 → 110 and enabled selective DEBUG logging—this increased sensitivity to brief overcurrent transients, exposing existing grid-side switching events as TRIP-31 protections.
2. Cluster events correlated temporally with utility capacitor bank switching (documented at 04:18Z at a nearby substation).

3. Concurrent update load on some gateways increased latency, causing transient telemetry gaps and escalated protective triggers on marginal devices.

### 5.4 Corrective & Preventative Actions

- Applied hotfix v3.15.0-hf1 adjusting sensing calibration and applied device-specific trip31_pct = 112 for affected devices.
- Reduced firmware_update.concurrent_sessions to 6 fleet-wide during rollouts to lower memory/latency pressure.
- Implemented pre-rollout grid-event watch: coordinate with TO to avoid rolling during scheduled switching windows.
- Enhanced post-upgrade monitoring dashboards to flag TRIP-31 clusters within 30 minutes and auto-escalate to vendor L2.

### 5.5 Action Items

| Action | Owner | Due |
|---|---|---|
| Inventory and mark devices that required hotfix | Firmware Team | 2025-12-15 |
| Update SOP with check for TO scheduled operations during rollouts | Ops Manager | 2025-12-10 |
| Implement automated rollback playbook integration to CMMS | Reliability Eng | 2026-01-05 |

## 6. Dispatch Runbooks & SCADA→CMMS Mappings

Runbooks map SCADA alarm codes and events to CMMS corrective work order templates. Use these templates to create standardized corrective actions and expedite vendor engagement.

### 6.1 Alarm → Work Order Mapping

| SCADA Alarm | Description | CMMS Template | Priority |
|---|---|---|---|
| scada_alarm_code:31 | TRIP-31 Overcurrent trip detected | WO-T-OC-31: "Investigate Overcurrent Trip" | High |
| scada_alarm_code:44 | TRIP-44 DC string/arc event | WO-T-DC-44: "String/Arc Inspection & Corrective" | Critical |
| firmware_update.failed | Firmware update failed | WO-T-FW-FAIL: "Firmware Failure Diagnostic" | Medium |

| | | | |
|---|---|---|---|
| device.reboot_loop | Device stuck in reboot loop | WO-T-REBOOTLOOP: "Device Replacement/ Recovery" | High |

### 6.2 Work Order Template — Example: WO-T-OC-31

Fields and example text used when creating the corrective work order.

```
Work Order ID: auto-generated
Template: WO-T-OC-31
Priority: High
Site: {plant_id}
Devices: {device_list}
Reported Time: {timestamp}
Symptoms: Multiple TRIP-31 events observed post-firmware upgrade;
inverter(s) in FAULT state.
Immediate Actions:
  - Remote reboot inverter(s)
  - Obtain power curve window t-30m..t+30m
  - Check protection.last_trip_code and last_trip_ts
Next Steps:
  - Schedule on-site technican if remote remediation fails
  - If >3 devices affected, trigger fleet escalation and consider
rollback
Attachments: Diagnostic bundle, SCADA event logs, vendor debug
output
```

### 6.3 Dispatch Playbook (If TRIP-31 cluster)

1. Create WO-T-OC-31 via CMMS and assign to plant crew.
2. Run remote commands: collect power_curve, protection logs, and device_config.
3. If remote reboot does not clear fault within 15 minutes, dispatch field tech with spare inverter controller.
4. Notify vendor L2 if multiple devices at same plant exhibit same failure mode.

## 7. Settlement & Imbalance Charge Guidance (ERCOT)

How curtailment and availability impacts are reconciled in ERCOT and which fields should be reconciled by the settlement team to minimize charges.

### 7.1 Key Concepts

- Settlement Interval: 15-minute intervals used by ERCOT.
- Availability: % of rated capacity available for settlement per interval.
- Imbalance Charges: Calculated when actual generation differs from

scheduled generation; may include Uplift and Ancillary charges.

### 7.2 Fields to Reconcile

| Field Name | Description / How to use |
|---|---|
| settlement_interval_ts | Start timestamp of the 15-minute interval (UTC) |
| metered_energy_MWh | MWh measured by revenue meter for interval |
| scheduled_dispatch_MW | ISO schedule in MW used for settlement |
| availability_percent | Calculated % availability for the interval (for outage adjustment) |
| curtailment_code | OPA curtailment code used when instructed by ISO/TO |
| event_reference_id | Internal incident/ticket ID to correlate deviations |

### 7.3 Example Imbalance Calculation (Simplified)

```
# Simplified per-interval imbalance (MWh)
imbalance_MWh = scheduled_dispatch_MW * 0.25 - metered_energy_MWh

# Penalty if imbalance > threshold (example)
if abs(imbalance_MWh) > 0.05:
    penalty = abs(imbalance_MWh) * imbalance_price ($/MWh)
else:
    penalty = 0.0
```

Operational recommendation: For intervals impacted by firmware-induced unavailability, tag settlement_interval entries with event_reference_id and include diagnostic proof (device logs, vendor ticket) when submitting disputes with ERCOT. This improves chances of remediation or reduced penalties.

### 7.4 Data Reconciliation Checklist

1. Confirm meter timestamp synchronization (NTP) and meter health.
2. Reconcile CMS availability vs vendor-reported device availability.
3. Attach device log bundle showing firmware_update timestamps aligned to settlement intervals for dispute support.

## 8. Contacts, Escalation Notes, and Ticket Comments

Consolidated contact points and escalations captured during the rollout. Use these details when searching the RAG index for human contacts and vendor responsibilities.

### 8.1 Contact Directory (primary)

| Role | Name / Team | Contact |
|---|---|---|
| Vendor L1 Support | Vendor Support Team | vendor-l1@vendor.com / +1-800-555-0101 |
| Vendor L2 Engineering | Vendor Eng | vendor-l2@vendor.com / +1-800-555-0102 |
| Vendor Escalation | Vendor Escalate | vendor-escalate@vendor.com / +1-800-555-0199 |
| Internal Operations On-call | Ops Oncall | ops-oncall@company.com / +1-555-1000 |
| Settlement Team | Settlements | settlements@company.com |

### 8.2 Escalation Notes (vendor)

Vendor escalation guidance captured in tickets:

- Escalate to L2 if TRIP-31 occurs > 2 times per site within 4 hours.
- Escalate to vendor-acct-manager if any rollback is performed or if portfolio availability decline > 1% impacts settlement.

### 8.3 Representative Time-Stamped Ticket Comments (for indexing)

```
# VEND-2025-11120-002 (selected)
2025-11-20T10:25Z - ops: "Riverbend: TRIP-31 at 10:12Z on
XG440-1000723; IVR shows FAULT."
2025-11-20T10:40Z - vendor-l2: "Requesting powercurve t-10m..t+10m
for 1000723 and protection logs."
2025-11-21T04:20Z - vendor-l2: "Multiple TRIP-31 events correlate
with local capacitor switching at substation; recommend HOLD."
2025-12-08T02:05Z - vendor: "Hotfix rollout initiated; target list
attached."
```

## 9. Appendix: Field Names, Error Codes, Example Logs

### 9.1 Canonical Field Names

| Field | Description / Example |
|---|---|
| device.device_id | Unique device identifier (e.g., XG440-1000723) |
| device.ip | Device management IP address |

| | |
|---|---|
| firmware_version | Installed firmware string (e.g., "3.15.0" or "3.15.0-hf1") |
| protection.last_trip_code | Numeric trip code last seen (e.g., 31) |
| protection.last_trip_ts | Timestamp of last trip event |
| scada_alarm_code | Aggregated code usable in dashboards for alarm counts |
| firmware_update.status | queued, downloading, installing, rebooting, online, failed |
| device_config.override | Boolean/JSON holding override parameters |

### 9.2 Error & Event Codes

| Code | Name | Meaning / Troubleshooting |
|---|---|---|
| TRIP-31 | Overcurrent Trip | Immediate shutdown; check protection thresholds and grid currents. Collect power curve and RMS samples. |
| TRIP-44 | DC String/Arc | Potential physical string fault. Site inspection required. Check I/R values. |
| FW-ERR-07 | Firmware apply checksum mismatch | Retry update; if persists, mark for vendor re-image. |
| COMM-LOST | Telemetry gap | Check gateway connectivity, packet loss metrics, and retry telemetry collection. |

### 9.3 Example Diagnostic Log Snippets

```
2025-11-21T02:09:58Z device:XG440-1001122 protection:
last_trip_code=31 last_trip_ts=2025-11-21T02:09:55Z
reason="Overcurrent instantaneous"
2025-11-21T02:10:02Z device:XG440-1001122 firmware_update:
status=online firmware_version=3.15.0
2025-11-21T02:10:10Z scada:alarm scada_alarm_code=31 plant=PLANT-
SC-010
```

### 9.4 Example SQL Queries for RAG Indexing

These example queries can be used to extract events, tickets, and logs for ingestion into a RAG index.

```
-- Query: devices that tripped with TRIP-31 during rollout window
```

```
SELECT device_id, protection.last_trip_ts, firmware_version,
plant_id
FROM devices
WHERE protection.last_trip_code = 31
  AND protection.last_trip_ts BETWEEN '2025-11-20T00:00:00Z' AND
'2025-11-22T00:00:00Z';

-- Query: tickets mentioning TRIP-31
SELECT ticket_id, created_ts, comments
FROM vendor_tickets
WHERE comments LIKE '%TRIP-31%'
  AND created_ts BETWEEN '2025-11-20' AND '2025-12-10';
```

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Notes: This document is intended for internal operations and knowledge-base ingestion. For real-world actions, refer to vendor support contracts and safety procedures.